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Some Myths About Bias: A Queer Studies Reading Of Gender Bias In NLP



(/pdf?id=gbAkGd3Ctd)

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Published: 30 May 2025, Last Modified: 09 Jun 2025 GeBNLP 2025

GeBNLP, Area Chairs, Reviewers, Authors Revisions (/revisions?id=gbAkGd3Ctd)

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Keywords: Gender, gender bias, gender binary, queer studies, bias evaluation and mitigation

TL;DR: This paper deploys theories about binary structures from Queer Studies to analyze assumptions about gender bias in NLP research.

Abstract:

This paper critiques common assumptions about gender bias in NLP, focusing primarily on word vector-based methods for detecting and mitigating bias. It argues that these methods assume a kind of "binary thinking" that goes beyond the gender binary toward a conceptual model that structures and limits the effectiveness of these techniques. Drawing its critique from the Humanities field of Queer Studies, this paper demonstrates that binary thinking drives two "myths" in gender bias research: first, that bias is categorical, measuring bias in terms of presence/absence, and second, that it is zero-sum, where the relations between genders are idealized as symmetrical. Due to their use of binary thinking, each of these myths flattens bias into a measure that cannot distinguish between the types of bias and their effects in language. The paper concludes by briefly pointing to methods that resist binary thinking, such as those that diversify and amplify gender expressions.

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Paper Decision

Decision by Program Chairs 📅 28 May 2025, 06:34 (modified: 30 May 2025, 10:07)

👁 Program Chairs, Area Chairs, Authors 📄 Revisions (/revisions?id=DPCJ1qb4dP)

Decision: Accept

Comment:

The paper offers a valuable critique of the prevalent binary framing of gender bias in NLP.

While the contribution is both timely and important, I believe it would be strengthened by acknowledging and citing the existing, albeit limited, body of work on biases affecting non-binary individuals.

Here are some examples:

- Gender Identity in Pretrained Language Models: An Inclusive Approach to Data Creation and Probing (Knupleš et al., Findings 2024)
- A Tale of Pronouns: Interpretability Informs Gender Bias Mitigation for Fairer Instruction-Tuned Machine Translation (Attanasio et al., EMNLP 2023)
- What about "em"? How Commercial Machine Translation Fails to Handle (Neo-)Pronouns (Lauscher et al., ACL 2023)

Official Review of Submission42 by Reviewer najN

Official Review by Reviewer najN 📅 20 May 2025, 16:19 (modified: 30 May 2025, 10:07)

👁 Program Chairs, Area Chairs, Reviewers Submitted, Reviewer najN, Authors

📄 Revisions (/revisions?id=4dTtRxo2uE)

Paper Summary:

This paper takes a critical look at some of the theoretical underpinnings of the conceptualization and operationalization of gender bias in NLP. It is clearly written and clearly articulates how two key myths (bias is categorical, and bias is zero-sum) arise from the binary conceptualization of bias *in general* as a dichotomy, as well as how these myths lead to major holes in the field. A few pointers towards methods that avoid these pitfalls are also suggested.

Summary Of Strengths:

Extremely clear and lays out the theory it relies on in a way that I think is quite friendly to an NLP audience, which is great because I think this is a super valuable paper for us who are practitioners in NLP to engage with. The two myths are well-explained and framed, with illustrative examples walking the reader through how particular methods and metrics come to contribute to these myths as well as the blind spots they produce. I particularly appreciate how issues such as the power structures linked to "opposing" terms being assumed as symmetrical are touched on without losing the thread of why this ends up producing problems that will be seen as "practical" in an NLP/computer science space rather than "just theory." I loved this paper and I wish you could see all the hearts and "!!yesssss"es that I have adorned it with while reviewing.

Summary Of Weaknesses:

There were two places I had some minor concerns about a claim, but I think they could pretty easily be fixed with clarification and maybe some extra citations:

1. on line 111 "[...] bias is always harmful" -- I'm not sure one can locate these myths as "driving" this particular assumption, because the assumption itself is a bit circular: bias in "algorithmic social bias" is *per definition* harmful, literally: as the term is used in this field biased algorithmic behavior is "harmful" algorithmic behavior (well-illustrated by the Barocas et al. paper you cite). I agree that it's problematic that this usage than exists alongside others (e.g. bias as any statistical skew; the many ways taxonomies of bias break down and harms get conflated), but this assumption (bias is harmful -> that's bad -> we should get rid of it) is often actually made pretty explicit and is itself a load-bearing reason to why it's treated as zero-sum, the snake eats its own tail, etc.
2. at the beginning of section 4 the paper mentions that the aim of bias detection and mitigation is "elimination of bias" and implies that within the field there aren't any works problematizing this, but in my experience it's definitely discussed (no silver bullets, the idea of "unbiased" as both ill-defined and a moving target) even in works you've already cited. I would politely suggest rephrasing this to show how the *extent* of problematizing this in Queer Studies is 1) obviously much stronger and 2) that's going to be useful for the NLP side so that we don't have to reinvent the wheel :)

Suggestions And Typos:

- Since you focus on word-vector techniques it might be worth clarifying what sub-class of "LLMs" you mean (i.e. not GPT-style) since that term is moving Really Fast right now in terms of what is "large" and I at least didn't immediately realize you were focused on methods for word-embedding/BERT-style LLMs and was expecting more references to QA tasks etc.
- I get why you use the Nemani et al. organization of "bias" categories but since you do also mention the representation/allocation divide it's probably worth mentioning that under this paradigm you're focused on representation (which in turn is often held to affect allocation...) side and that's why these are the three you're taking for your framework. It's kind of gotten at but quite late in the paper so I had found myself wondering
- l. 207 stray "[" at the Devinney et al. citation
- l.313 "type of bias, to denigration, which *either* ignores..." there's no 'or' later in this sentence, is 'either' a typo?
- l. 552 "harness" -> "harnesses" ?

Bias Statement: Yes

Overall Score: 5: Top-Notch: This is one of the best papers I read recently, of great interest for the (broad or narrow) sub-communities that might build on it

Confidence: 5: Positive that my evaluation is correct. I read the paper very carefully and am familiar with related work.

Meta Review of Submission42 by Area Chair wJA2

Meta Review by Area Chair wJA2 📅 19 May 2025, 16:29 (modified: 30 May 2025, 10:07)

👁 Area Chairs, Authors, Reviewers Submitted, Program Chairs

📄 Revisions (/revisions?id=57xTta0AFW)

Metareview:

This paper gives a fresh point of view and contributions to NLP biases studies. The authors should address minor issues raised by reviewers.

Overall Recommendation: Strong Accept

Confidence: 4: The area chair is confident but not absolutely certain

Official Review of Submission42 by Reviewer NTgr

Official Review by Reviewer NTgr 📅 19 May 2025, 13:08 (modified: 30 May 2025, 10:07)

👁 Program Chairs, Area Chairs, Reviewers Submitted, Reviewer NTgr, Authors

📄 Revisions (/revisions?id=5lGenQNNTP)

Paper Summary:

This paper critiques mainstream approaches to detecting and mitigating gender bias in NLP by applying insights from Queer Studies. It challenges two dominant "myths" in gender bias research:

Bias is Categorical – Bias is treated as either present or absent. Bias is Zero-Sum – Bias is assumed to be symmetrical between genders. The authors argue that these assumptions stem from binary thinking, which oversimplifies the complex, fluid, and intersectional nature of gender. The paper emphasizes that bias is not an anomaly to be removed but a structural feature of language shaped by power and ideology.

Summary Of Strengths:

- The paper is grounded in rich theoretical traditions from Queer Studies, Feminist Theory, and Intersectionality, offering a fresh and critical lens on NLP practices.
- By bridging humanities and computational linguistics, the paper encourages more inclusive and reflective AI development.

Summary Of Weaknesses:

While the critique is strong, the paper offers limited practical guidance or empirical methods for implementing non-binary or intersectional bias detection.

Suggestions And Typos:

The writing could be improved by not mentioning "I" in the statements/evidences - but to be more general ("in this paper", "from the experiments", etc.)

Bias Statement: Yes

Overall Score: 3.5

Confidence: 4: Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Official Review of Submission42 by Reviewer FKS1

Official Review by Reviewer FKS1 📅 01 May 2025, 10:05 (modified: 30 May 2025, 10:07)

👁 Program Chairs, Area Chairs, Reviewers Submitted, Reviewer FKS1, Authors

📄 Revisions (/revisions?id=K0Dk9nsSBk)

Paper Summary:

This interdisciplinary paper critiques prevailing assumptions in NLP gender bias research through the lens of Queer Studies. It argues that many bias detection and mitigation methods (e.g., WEAT, DeBias, gender-swapping techniques) are constrained by "binary thinking," which oversimplifies bias into categorical (present/absent) or zero-sum frameworks. Drawing on Queer theorists, the authors demonstrate how binaries obscure power dynamics and diverse bias types (denigration, stereotyping, underrepresentation). They propose alternatives like bias amplification and "gender geometries" to disrupt binary limitations and encourage richer, more equitable representations.

Summary Of Strengths:

- Critical and detailed analysis of previous related papers based on the two mentioned assumptions

(categorical and zero-sum frameworks)

- Integrates humanities-based critiques (Queer Studies) into NLP, offering a fresh theoretical lens to interrogate bias methodologies.
- Adopts Nemani et al.'s tripartite framework (denigration, stereotype, underrepresentation) to disentangle conflated bias types in prior work.

Summary Of Weaknesses:

- Other bias-mitigation paradigms, besides word-vector and prompt methods, are only briefly mentioned or omitted.
- While the paper presents theoretical insights, the concrete steps researchers should take to implement these insights in practice remain somewhat vague. More specific guidelines for developing bias detection and mitigation techniques based on the Queer Studies perspective would strengthen the paper.

Suggestions And Typos:

- In line 571, "journies" should be "journeys". Also, in line 146, "particular kind the implication" should be "particular kind of implication".
- Add a glossary or more extensive explanations of key Queer Studies concepts for NLP researchers unfamiliar with this literature. Apart from that, the author could add more details beyond theoretical aspects by providing steps to implement the suggested solution.

Bias Statement: Yes

Overall Score: 3.5

Confidence: 3: Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

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